# Discrete Probability Theory
# revision course

Jonas Hübotter

# Outline

Counting

Probability

Conditional probability

Discrete random variables

Continuous random variables

Inductive Statistics

Markov chains

# Plan I

Counting
- Sample spaces and events
- Counting sets

# Sample spaces and events

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Assumptions:

- all outcomes equally likely
- finite sample space

# Counting sets

## Multiplication rule

Consider $i \in [m]$ experiments with $n_i$ possible outcomes. Then the overall number of possible outcomes is

$$\prod_{i=1}^{m} n_i.$$

## Sampling table

Given $n$ objects, select $k$ objects.

| | order | $\neg$ order |
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# Plan I

Probability
- $\sigma$-algebras
- Probability spaces
- Joint and marginal probabilities

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### Why do we need $\sigma$-algebras?

To describe events in the context of a probability space.

# Probability spaces

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- $P(S) = 1$;
- $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ if $\forall i \neq j.\ A_i \cap A_j = \emptyset$.

### Definition 5

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### Definition 6

A probability space consists of

- a sample space $S$;
- a $\sigma$-algebra $\mathcal{A}$ over $S$; and
- a probability measure $P$ on $\mathcal{A}$.

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- if $A, B \in \mathcal{A}$ and $A \subseteq B$, then $P(A) \leq P(B)$

Also the principle of inclusion-exclusion holds:

$$P(\bigcup_{i=1}^{n} A_i) = \sum_{I \subseteq [n], I \neq \emptyset} (-1)^{|I|+1} \cdot P(\bigcap_{i \in I} A_i).$$

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And Boole's inequality holds:

$$P(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i).$$

# Joint and marginal probabilities

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A joint probability is the probability of two or more events occurring simultaneously:

$$P(A, B) = P(A \cap B).$$

# Plan I

Conditional probability
- Prior and posterior
- Independence
- Conditioning

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$$P(A|B) = \frac{P(A, B)}{P(B)}.$$

The posterior is the joint probability of the event $A$ and the information $B$ relative to the probability of the information $B$.

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- $P(A, B) = P(B)P(A|B) = P(A)P(B|A)$
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- $P(A_1, \ldots, A_n) =$
  $P(A_1)P(A_2|A_1)P(A_3|A_1, A_2) \cdots P(A_n|A_1, \ldots, A_{n-1})$
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  (multiplication rule)
- $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ (Bayes' rule)
- $P(A) = P(A, B) + P(A, \bar{B}) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$
  (law of total probability)

# Plan I

Discrete random variables
- Cumulative Distribution Function
- Probability Mass Function
- Independence
- Bernoulli
- Averages
- Indicator variables
- Binomial
- Variance
- Geometric
- Poisson
- Probability-generating functions
- Moment-generating functions
- Joint distributions
- Conditional distributions

# Plan II

Convolutions
More distributions
Inequalities

# Discrete random variables

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The range of a discrete random variable

$$X(S) = \{x \in \mathbb{R}.\ \exists A \in S.\ X(A) = x\}$$

is also discrete.

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- $F_X(x) \xrightarrow{x \to -\infty} 0$
- $F_X(x) \xrightarrow{x \to \infty} 1$

Therefore, $P(a < X \leq b) = F_X(b) - F_X(a)$.

# Probability Mass Function

Definition 9

The probability mass function of a discrete random variable $X$ is defined as $f_X(x) = P(X = x) \in [0, 1]$ where

$$\sum_{x \in X(S)} f_X(x) = 1.$$

The CDF of $X$ can be obtained from the PDF of $X$ by summing over the PDF

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The PMF of $X$ can be obtained from the CDF of $X$ by identifying the *jumps* in the CDF

$$f_X(x) = F_X(x) - F_X(prev(x)).$$

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# Bernoulli

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**Overview**

- $E(X) = p$
- $Var(X) = p(1 - p)$
- $G_X(s) = 1 - p + ps$
- $M_X(s) = 1 - p + pe^s$

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For infinite probability spaces absolute convergence of $E(X)$ is necessary for the existence of $E(X)$.

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- $E(a \cdot X + b) = a \cdot E(X) + b$, $E(X + Y) = E(X) + E(Y)$ (linearity)
- $E(\prod_{i=1}^{n} X_i) = \prod_{i=1}^{n} E(X_i)$ if $X_1, \ldots, X_n$ independent (multiplicativity).

Definition 12

$E(X^i)$ is called the *i*-th moment of the random variable $X$ and $E((X - E(X))^i)$ is called the *i*-th central moment of $X$.

The law of the unconscious statistician (LOTUS) can be used to find the expected value of transformed random variables.

$$E(g(X)) = \sum_{x \in X(S)} g(x) \cdot P(X = x).$$

# Indicator variables

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Properties of indicator variables:

- $E(I_A) = P(A)$ (fundamental bridge)
- $E(I_{A_1} \cdots I_{A_n}) = P(A_1 \cap \cdots \cap A_n)$.

# Binomial

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- $Var(X) = np(1-p)$
- $G_X(s) = (1-p+ps)^n$
- $M_X(s) = (1-p+pe^s)^n$

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- $Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$ if $X_1, \ldots, X_n$ independent.

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- $E(X) = \frac{1}{p}$

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- $E(X) = \frac{1}{p}$
- $Var(X) = \frac{1-p}{p^2}$

# Geometric

### Definition 16 ($X \sim Geom(p)$)

A discrete random variable $X$ has the geometric distribution with parameter $p$ when $X$ models the #trials leading up to a success in independent $Bern(p)$ trials.

$$f_X(k) = p(1-p)^{k-1}, k \in \mathbb{N}. \qquad F_X(k) = 1 - (1-p)^{\lfloor k \rfloor}.$$

### Overview

- $E(X) = \frac{1}{p}$
- $Var(X) = \frac{1-p}{p^2}$
- $G_X(s) = \frac{ps}{1-(1-p)s}$

## Memorylessness

Completing $x$ trials that are all failures does not change the probability of the next $y$ trials to include a success.

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The geometric distribution is the only memoryless discrete distribution.

# Poisson

### Definition 17 ($X \sim Po(\lambda)$)

A discrete random variable $X$ has the Poisson distribution with parameter $\lambda$ when $X$ models the #events in a fixed interval with rate $\lambda$ and with events independently occurring of the time since the last event.

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### Overview

- $E(X) = \lambda$

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### Overview

- $E(X) = \lambda$
- $Var(X) = \lambda$
- $G_X(s) = exp(\lambda(s-1))$
- $M_X(s) = exp(\lambda(e^s - 1))$

## Poisson approximation to the Binomial

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Then the distribution of $X$ converges to $Po(\lambda)$ as $n \to \infty$ (i.e. for small $\lambda/n$).

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### Definition 18

Given a discrete random variable $X$ with $X(S) \subseteq \mathbb{N}_0$ the probability-generating function is defined as

$$G_X(s) = \sum_{x \in X(S)} s^x \cdot P(X = x)$$

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The PGF of a random variable $X$ generates the PMF of $X$:

$$P(X = i) = \frac{G_X^{(i)}(0)}{i!}.$$

Properties of probability-generating functions:

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- $E(X) = G'_X(1)$

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- $G_{X+Y}(s) = G_X(s) \cdot G_Y(s)$ if $X, Y$ independent
- $G_Z(s) = G_N(G_X(s))$ for $Z = X_1 + \cdots + X_N$, $X_i$ i.i.d. with PGF $G_X$, and $N$ independent.

# Moment-generating functions

## Definition 19

Given a random variable $X$ the moment-generating function is defined as

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$$\begin{aligned} M_X(s) &= \sum_{x \in X(S)} e^{sx} \cdot P(X = x) \\ &= E(e^{sX}) \\ &= \sum_{i=0}^{\infty} \frac{E(X^i)}{i!} \cdot s^i. \end{aligned}$$

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Given a random variable $X$ the moment-generating function is defined as

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The MGF of a random variable $X$ generates the $i$-th moment of $X$:

$$E(X^i) = M_X^{(i)}(0).$$

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- $M_X(s) = G_X(e^s)$ if $X(S) \subseteq \mathbb{N}_0$

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- $M_X(s) = G_X(e^s)$ if $X(S) \subseteq \mathbb{N}_0$
- $M_{X+Y}(s) = M_X(s) \cdot M_Y(s)$ if $X, Y$ independent.

# Joint distributions

### Definition 20

A joint distribution is the distribution of two or more random variables.

$$f_{X,Y}(x, y) = P(X = x, Y = y).$$

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The marginal distribution of a random variable can be obtained from a joint distribution by summing over all other random variables:

$$f_X(x) = \sum_{y \in Y(S)} f_{X,Y}(x, y).$$

# Conditional distributions

### Definition 21

Given the joint distribution of two random variables $X$ and $Y$ the conditional distribution of $X$ given $Y$ is the distribution of $X$ when $Y$ is known to be a particular value.

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_{Y|X}(y|x) \cdot f_X(x)}{f_Y(y)}.$$

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The conditional expectation of the random variables $X|Y = y$ is the expected value of the distribution $f_{X|Y=y}$:

$$E(X|Y = y) = \sum_{x \in X(S)} x \cdot f_{X|Y}(x|y).$$

# Convolutions

**Definition 22**

Let $X$ and $Y$ be independent and $Z = X + Y$. Then

$$f_Z(z) = \sum_{x \in X(S)} f_X(x) \cdot f_Y(z - x).$$

# Convolutions

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Let $X$ and $Y$ be independent and $Z = X + Y$. Then

$$f_Z(z) = \sum_{x \in X(S)} f_X(x) \cdot f_Y(z - x).$$

The derivation of the distribution of a sum of random variables given the marginal distributions is called convolution.

# More distributions

Definition 23 ($X \sim HypGeom(r, a, b)$)

A discrete random variable $X$ has the hypergeometric distribution with parameters $r, a$ and $b$ when $X$ models the # of drawn objects that have a specified feature in $r$ draws without replacement from $a + b$ objects where $b$ objects have the specified feature.

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$$f_X(x) = \frac{\binom{b}{x}\binom{a}{r-x}}{\binom{a+b}{r}}.$$

# More distributions

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$$f_X(x) = \frac{\binom{b}{x}\binom{a}{r-x}}{\binom{a+b}{r}}.$$

Overview

- $E(X) = r \cdot \frac{b}{a+b}$

Definition 24 ($Z \sim NegBin(n, p)$)

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$$f_Z(z) = \binom{z-1}{n-1} p^n (1-p)^{z-n}.$$

### Example 25

Let $X_1, \ldots, X_n \sim Geom(p)$ i.i.d.
Then $Z = X_1 + \cdots + X_n \sim NegBin(n, p)$.

# Inequalities

## Inequalities vs approximations

*Approximations* allow us to model more complex problems but you usually don't know how good the approximation is.

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*Approximations* allow us to model more complex problems but you usually don't know how good the approximation is.
*Inequalities* allow us to prove definite facts (i.e. bounds) about probabilities of certain events.

### Definition 26 (Markov)

Given a random variable $X \geq 0$ and $t > 0$

$$P(X \geq t) \leq \frac{E(X)}{t}.$$

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### Definition 27 (Chebyshev)

Given a random variable $X$ and $t > 0$

$$P(|X - E(X)| \geq t) \leq \frac{Var(X)}{t^2}.$$

### Definition 28 (Chernoff)

Let $X_1, \ldots, X_n$ be independent, Bernoulli-distributed random variables with $X_i \sim Bern(p_i)$. Then the following inequalities hold for $X = \sum_{i=1}^{n} X_i$ and $\mu = E(X) = \sum_{i=1}^{n} p_i$.

## Definition 28 (Chernoff)

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- $P(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$ for all $\delta > 0$;
- $P(X \leq (1-\delta)\mu) \leq \left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\mu}$ for all $0 < \delta < 1$

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- $P(X \geq (1+\delta)\mu) \leq e^{-\mu\delta^2/3}$ for all $0 < \delta \leq 1$;
- $P(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}$ for all $0 < \delta \leq 1$;
- $P(|X - \mu| \geq \delta\mu) \leq 2e^{-\mu\delta^2/3}$ for all $0 < \delta \leq 1$;
- $P(X \geq (1+\delta)\mu) \leq \left(\frac{e}{1+\delta}\right)^{(1+\delta)\mu}$; and
- $P(X \geq t) \leq 2^{-t}$ for all $t \geq 2e\mu$.

# Plan I

Continuous random variables
- Measure Theory
- Continuous probability spaces
- Uniform
- Normal (Gaussian)
- $\gamma$-quantiles
- Exponential
- Joint distributions
- More distributions

# Continuous random variables

Definition 29

A continuous random variable $X$ is a function

$$X : S \to \mathbb{R}$$

where $X(S)$ is uncountable.

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$$X : S \to \mathbb{R}$$

where $X(S)$ is uncountable.

The distribution of $X$ is defined by the probability density function $f_X : \mathbb{R} \to \mathbb{R}_0^+$ with the property

$$\int_{-\infty}^{+\infty} f_X(x)\ dx = 1.$$

# Measure Theory

## Definition 30

- A Borel set of $\mathbb{R}$ is any subset $A \subseteq \mathbb{R}$ which can be represented as countably many unions and intersections of intervals (open, half-open, or closed) on $\mathbb{R}$.

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- For a measurable function $f$ we denote the Lebesgue integral by $\int f \; d\lambda$.

## Example 31 (Examples of measurable functions)

- the characteristic function $1_A$ of the set $A$,
- continuous functions, and
- sums and products of measurable functions.

## Probability spaces over Borel sets

The set of Borel sets $\mathcal{A}$ is a $\sigma$-algebra over $\mathbb{R}$.

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Especially, $P$ satisfies the Kolmogorov axioms.

# Continuous probability spaces

### Definition 32

An event is a set $A = \bigcup_k I_k \subseteq \mathbb{R}$ that can be resembled as the union of countably many pairwise disjunct intervals. The probability of $A$ is given as

$$P(A) = \int_A f_X(x)\ dx = \sum_k \int_{I_k} f_X(x)\ dx.$$

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The probability of the event $A = \{x\}, x \in \mathbb{R}$ is always 0.

## Cumulative distribution functions

The cumulative distribution function of a continuous random variable $X$ is given as

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The PDF of $X$ can be obtained from the CDF of $X$ by finding its derivative with respect to $x$:

$$f_X(x) = \frac{dF_X}{dx}.$$

## Intervals

By the fundamental theorem of calculus, the probability of $X$ being in the interval between $a$ and $b$ is given as

$$P(a \leq X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\ dx.$$

## Expected values

The expected value of a continuous random variable $X$ is given as

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The law of the unconscious statistician still holds in the continuous case:

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x) \cdot f_X(x)\ dx.$$

# Uniform

**Definition 33 ($X \sim Unif(a, b)$)**

A continuous random variable $X$ is uniformly distributed with parameters $a$ and $b$ when $X$ models the outcome of an experiment where all outcomes that lie in the interval $[a, b]$ are equally likely.

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Overview

- $E(X) = \frac{a+b}{2}$

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Overview

- $E(X) = \frac{a+b}{2}$
- $Var(X) = \frac{(a-b)^2}{12}$

## Universality of the Uniform

Let $X \sim F$. Then $F(X) \sim Unif(0, 1)$.

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Let $X \sim F$. Then $F(X) \sim Unif(0, 1)$.

Realizations of a random variable of any distribution $F$ with the inverse CDF $F^{-1}$ can be simulated using realizations of a uniformly distributed random variable $Y$: $F^{-1}(Y) \sim F$.

# Normal (Gaussian)

Definition 34 ($X \sim \mathcal{N}(\mu, \sigma^2)$)

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) =: \varphi(x; \mu, \sigma).$$

$$F_X(x) =: \Phi(x; \mu, \sigma).$$

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Overview

- $E(X) = \mu$

# Normal (Gaussian)

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Overview

- $E(X) = \mu$
- $Var(X) = \sigma^2$

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$$F_X(x) =: \Phi(x; \mu, \sigma).$$

Overview

- $E(X) = \mu$
- $Var(X) = \sigma^2$
- $M_Z(s) = exp(\mu s + \frac{(\sigma s)^2}{2})$

$\mathcal{N}(0, 1)$ is the standard normal distribution.

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### Linear transformation

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then for any $a \in \mathbb{R} \setminus \{0\}$ and $b \in \mathbb{R}$ the random variable

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is normally distributed with mean $a\mu + b$ and variance $a^2\sigma^2$.

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### Standardization

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = \frac{X-\mu}{\sigma}$. Then $Y \sim \mathcal{N}(0,1)$.
The random variable $Y$ is called standardized.

## Additivity

Let $X_1, \ldots, X_n$ independent and normally distributed with parameters $\mu_i, \sigma_i^2$. Then the random variable

$$Z = a_1 X_1 + \cdots + a_n X_n$$

is normally distributed with mean $a_1\mu_1 + \cdots a_n\mu_n$ and variance $a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2$.

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## Normal approximation to the Binomial

Let $X \sim Bin(n, p)$ with CDF $F_n(t)$. Then

$$F_n(t) \approx \Phi\left(\frac{t - np}{\sqrt{p(1-p)n}}\right)$$

can be used as an approximation if $np \geq 5$ and $n(1-p) \geq 5$.

# $\gamma$-quantiles

### Definition 35

Let $X$ be a continuous random variable with distribution $F_x$. A number $x_\gamma$ with

$$F_X(x_\gamma) = \gamma$$

is called $\gamma$-quantile of $X$ or the distribution $F_X$.

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### Definition 36

For the standard normal $z_\gamma$ denotes the $\gamma$-quantile.

# Exponential

**Definition 37 ($X \sim Exp(\lambda)$)**

A continuous random variable $X$ is exponentially distributed with parameter $\lambda$ when $X$ models the time between events in a Poisson process.

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- $Var(X) = \frac{1}{\lambda^2}$

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### Overview

- $E(X) = \frac{1}{\lambda}$
- $Var(X) = \frac{1}{\lambda^2}$
- $M_X(s) = \frac{\lambda}{\lambda - s}, s < \lambda$

## Scaling

Let $X \sim Exp(\lambda)$. If $a > 0$, then $Y = aX$ is exponentially distributed with the parameter $\lambda/a$.

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## Memorylessness

The exponential distribution is the only memoryless continuous distribution. Therefore, any continuous random variable $X$ where

$$P(X > y + x | X > x) = P(X > y)$$

holds for all $x, y > 0$ is exponentially distributed.

## Waiting for multiple events

Let $X_1, \ldots, X_n$ be independent, exponentially distributed random variables with parameters $\lambda_1, \ldots, \lambda_n$. Then $X = \min\{X_1, \ldots, X_n\}$ is exponentially distributed with parameter $\lambda_1 + \cdots + \lambda_n$.

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## Exponential approximation to the Geometric

Let $X_n \sim Geom(\lambda/n)$. The distribution of scaled geometrically distributed random variables $Y_n = \frac{1}{n} X_n$ converges to an exponential distribution with parameter $\lambda$ as $n \to \infty$.

## Poisson process

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For $t > 0$ we define

$$X(t) = \max\{n \in \mathbb{N} \mid T_1 + \cdots + T_n \leq t\}$$

resembling the number of events that occurred up until time $t$.

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resembling the number of events that occurred up until time $t$.
Then $X(t)$ is Poisson-distributed with parameter $t\lambda$.

# Joint distributions

## Getting marginals

Given a joint distribution $f_{X,Y}$ the marginal distribution $f_X$ can be obtained as follows:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y)\ dy.$$

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## Calculating probabilities

Given an event $A \in \mathbb{R}^2$ the probability of $A$ is the area under the probability density function of $X$:

$$P(A) = \iint_A f_{X,Y}(x, y)\ dx\ dy.$$

## Finding PDFs

Given a joint CDF $F_{X,Y}$ the joint PDF $f_{X,Y}$ can be obtained as follows:

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x,y).$$

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## Finding CDFs

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$$F_{X,Y}(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u,v)\ du\ dv.$$

# More distributions

Definition 38 ($X \sim Lognormal(\mu, \sigma^2)$)

A continuous random variable $X$ is logarithmically normal distributed with parameters $\mu$ and $\sigma^2$ when $Y = ln(X) \sim \mathcal{N}(\mu, \sigma^2)$.

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A continuous random variable $X$ is logarithmically normal distributed with parameters $\mu$ and $\sigma^2$ when $Y = ln(X) \sim \mathcal{N}(\mu, \sigma^2)$.

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \cdot exp\left(-\frac{(ln(x) - \mu)^2}{2\sigma^2}\right)$$

for $x > 0$.

# Plan I

Inductive Statistics
- Estimators
- Maximum likelihood estimators
- Law of Large Numbers
- Central Limit Theorem
- Confidence intervals
- Hypothesis tests
- Statistical tests

# Inductive Statistics

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To generate data $n$ independent copies of an identical experiment modeled by the random variable $X$ are conducted. A measurement resulting from one of these experiments is called a sample.
Each sample is represented by a separate random variable $X_i$ called sample variable.

# Estimators

### Definition 39

An estimator for parameter $\theta$ is a random variable composed of multiple sample variables used to estimate $\theta$.

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The bias of an estimator $U$ is given as $E(U - \theta)$.

An estimator $U$ is unbiased for the parameter $\theta$ if $E(U) = \theta$ (i.e. its bias is zero).

### Definition 40

The sample mean $\bar{X}$ is an unbiased estimator for $E(X)$.

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

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### Definition 41

The sample variance $S^2$ is an unbiased estimator for $Var(X)$.

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

Definition 42

The mean squared error is a qualitative measure of an estimator $U$.

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An estimator $U$ is consistent in mean square if $MSE(U) \xrightarrow{n \to \infty} 0$.

# Maximum likelihood estimators

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Given sample variables $\overrightarrow{X} = (X_1, \ldots, X_n)$ and samples $\overrightarrow{x} = (x_1, \ldots, x_n)$, find Maximum-Likelihood estimator for $X$ with parameter $\theta$.

1. construct $L(\overrightarrow{x}; \theta) = f_{\overrightarrow{X}}(\overrightarrow{x}; \theta) = \prod_{i=1}^{n} f_{X_i}(x_i; \theta)$, modeling the likelihood that the samples $\overrightarrow{x}$ are described by $\theta$
2. find $\theta$ maximizing $L$, or equivalently $\ln L(\overrightarrow{x}; \theta) = \sum_{i=1}^{n} \ln f_{X_i}(x_i; \theta)$
3. the value for $\theta$ maximizing $L$ is a Maximum-Likelihood estimator for $\theta$

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$$P(|\bar{X} - E(X)| \geq \delta) \leq \epsilon$$

for $\delta, \epsilon > 0$ and $n \geq \frac{Var(X)}{\epsilon\delta^2}$.

# Central Limit Theorem

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Equivalently:

$$\sqrt{n}\left(\frac{\bar{X} - \mu}{\sigma}\right) \xrightarrow{n\to\infty} \mathcal{N}(0,1) \text{ in distribution.}$$

## De Moivre-Laplace theorem

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Let $X_1, \ldots, X_n \sim Bern(p)$ i.i.d. and $H_n = X_1 + \cdots + X_n$. Then

$$H_n^* = \frac{H_n - np}{\sqrt{np(1-p)}} \xrightarrow{n \to \infty} \mathcal{N}(0,1) \text{ in distribution.}$$

# Confidence intervals

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$[U_1, U_2]$ is a confidence interval.

Often a single estimator $U$ is used to define the symmetrical confidence interval $[U - \delta, U + \delta]$.

# Hypothesis tests

Given sample variables $\overrightarrow{X} = (X_1, \ldots, X_n)$ and sample values $\overrightarrow{x} = (x_1, \ldots, x_n)$ decide whether to accept or reject a hypothesis.

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$K$ is constructed based on the concrete values of the test variable $T$ that is composed of the sample variables.

A test is called one-sided if $K$ is a half-open interval in $T(S)$ and two-sided if $K$ is a closed interval in $T(S)$.

$H_0$ is the hypothesis to be tested, also called null-hypothesis.
$H_1$ is the alternative. $H_1$ is trivial if it is just the negation of $H_0$.

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## Errors

- type 1 error or $\alpha$-error or significance level
  $H_0$ holds, but $\overrightarrow{x} \in K$

$$\alpha = \sup_{p \in H_0} P_p(T \in K).$$

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- type 1 error or $\alpha$-error or significance level
  $H_0$ holds, but $\overrightarrow{x} \in K$

$$\alpha = \sup_{p \in H_0} P_p(T \in K).$$

- type 2 error or $\beta$-error
  $H_1$ holds, but $\overrightarrow{x} \notin K$

$$\beta = \sup_{p \in H_1} P_p(T \notin K).$$

The quality function $g$ describes the probability that a test rejects the null-hypothesis.

$$g(p) = P_p(T \in K).$$

# Statistical tests

## Characteristics

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  - Independence of involved random variables
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  In case of a two sample test:
  - Independence of involved random variables
    Are independent measurements (independence) or related measurements (dependence) taken?
  - Relationships between several random variables
    Regression analysis describes the examination of functional dependencies between random variables, whereas dependency analysis describes the examination of random variables regarding on independence.

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  Which assumptions does the test make regarding independence, distribution, expected value or variance?

## Important statistical tests

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# Plan I

Markov chains
- Stochastic processes
- Markov property
- Representations
- Probabilities
- Hitting times
- Stationary distribution
- Interlude: Diagonalization
- Convergence
- Properties

# Stochastic processes

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A stochastic process is a sequence of random variables $(X_t)_{t \in T}$ that describe the behavior of a system at time $t$.

If $T = \mathbb{N}_0$, the stochastic process has discrete time. If $T = \mathbb{R}_0^+$, the stochastic process has continuous time.
If $X_t$ is discrete (i.e. its range is countable), the system is said to have a distinct state at time $t$.

# Markov property

### Definition 44

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This property can be formalized as follows:

$$P(X_{t+1} = j | X_t = i_t, \ldots, X_0 = i_0) = P(X_{t+1} = j | X_t = i_t)$$

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This property can be formalized as follows:

$$P(X_{t+1} = j | X_t = i_t, \ldots, X_0 = i_0) = P(X_{t+1} = j | X_t = i_t) =: p^t_{i_t j}.$$

### Definition 45

A (finite) Markov chain (with discrete time) over the state space $S = \{0, \ldots, n-1\}$ consists of an infinite sequence of random variables $(X_t)_{t \in \mathbb{N}_0}$ with codomain $S$

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A (finite) Markov chain (with discrete time) over the state space $S = \{0, \ldots, n-1\}$ consists of an infinite sequence of random variables $(X_t)_{t \in \mathbb{N}_0}$ with codomain $S$ and the initial distribution $q_0$ with $q_0^T \in \mathbb{R}^n$.

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Farther, the Markov property must hold.

# Representations

### Definition 46

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The transition diagram is a graph consisting of vertices $S$ and weighted edges represented by the adjacency matrix $P$.

A concrete instance of the system can be interpreted as a random walk on the transition diagram.

# Probabilities

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The entries of $P^k$ refer to the probability of transitioning from state $i$ to state $j$ in exactly $k$ steps:

$$p_{ij}^{(k)} = P(X_{t+k} = j | X_t = i) = (P^k)_{ij}.$$

# Hitting times

### Definition 48

The hitting time of state $j$ from state $i$ is modeled by the following random variable:

$$T_{ij} = \min\{n \geq 1 \mid X_n = j \text{ given } X_0 = i\}.$$

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The expected recurrence time $h_i = h_{ii}$ and the recurrence probability $f_i = f_{ii}$ are defined analogously to the expected hitting time and the arrival probability.

# Stationary distribution

### Definition 50

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A Markov chain does not necessarily converge to a stationary distribution. Convergence depends on the properties of the Markov chain itself and its initial distribution.

# Interlude: Diagonalization

For eigenvectors $x_i$ and related eigenvalues $\lambda_i$ of a matrix $A$, $A \cdot x_i = \lambda_i \cdot x_i$ holds.

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Then $V^{-1} \cdot A \cdot V = \Lambda$ is called diagonalization of $A$.
Conversely, $A = V \cdot \Lambda \cdot V^{-1}$ holds.

# Convergence

From the diagonalization of the transition matrix it follows that

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$$\lim_{t\to\infty} P(X_t = j \mid X_0 = i) = \lim_{t\to\infty} P^t(i,j).$$

# Properties

Certain properties of Markov chains allow us to draw conclusions about its stationary distributions.

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if conversely $f_i < 1$, the state $i$ is transient.

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The Markov chain does not necessarily converge to the stationary distribution (periodicity!).

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Then the period of state $i$ is defined as $d_i = gcd(T(i))$.

If a Markov chain is irreducible, all of its states share the same period. This period is then referred to as the period of the Markov chain.

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A Markov chain is aperiodic if all its states are aperiodic.

### Definition 55

An ireducible and aperiodic Markov chain is called ergodic.

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For every finite ergodic Markov chain it holds independently of its initial distribution $q_0$ that

$$\lim_{t \to \infty} q_t = \pi$$

where $\pi$ refers to its unique stationary distribution.

### Definition 56

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For every finite ergodic Markov chain whose transition matrix is doubly stochastic its unique stationary distribution assigns the same probability to each state:

$$\pi \equiv \frac{1}{|S|}.$$